Forensic Report

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Sam Houston State

Background: An ordinance prohibiting the possession of nine or more Rhinos of rhinos passed by The University of New Orleans' network administrator saw questionable network behavior. Three network trace queues and a USB Storage device taken from one of The university labs were given to The investigators. As a forensic disk image, the USB device was saved. This report's goal is to use Forensic Technologies to reconstruct The Suspect's Actions and examines digital data to ascertain whether the Suspicious shared or possessed illegal Rhinoceros photos.

Task 1: The first task required within this analysis is to look over a set of files referred to as RHINOUSB and its log files under rhino.log. While investigating, hidden images are the focal point.

Task 2: Open either wireshark or autopsy to begin analyzing these files. What you will find is the images in a form or digital data that must be processed.

Task 3: Depending on what software you chose the file analysis will vary. In wireshark you must follow slide stream data. Autopsy is more simple as all you need to do is open the files in the platform.

Hypothesis 1: While analyzing these files the suspect may have the files too strongly encrypted/hidden for the study to continue further. Causing the crime to be successful

Hypothesis 2: The suspect may have the files stored on an external drive that is inaccessible without a machine. Making the data unreachable unless said drive is confiscated.

Hypothesis 3: The suspect is bluffing by putting the bait images in this file but the truly illegal images are in another file system. As low as the odds are, there is still a possibility of this taking place.

Domain Terms:

Accounts Payable- Money owed to vendors or suppliers for items and services obtained but not yet paid for. It frequently refers to short-term obligations and is listed as a liability on the balance sheet of the business.

Accounts Receivable- Funds that clients owe an organization for provided goods or services but have yet to be paid for. It indicates anticipated incoming money and is marked as an asset.

CPA- CPA's are qualified to carry out forensic accounting, financial reporting, auditing, and tax preparation. This is a certified public accountant who has completed state certification requirements and passed the Uniform CPA Examination.

Acquired Data:

Filename: RHINOUSB.zip

Type: Disk Image

Contents: rhino.log/rhino2.log/rhino3.log

Software: Wireshark + Autopsy

Acquired on: 02/2026

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Suspect Information:

Name- Iaman Informant

Threat Type- Insider (likely driven by anger/frustration)

Sophistication- Intermediate level threat

With viewing how suspects have carried themselves within the crime, there is a trail of motives fueled by anger, and or a feeling of revenge. Further leading to the crime being fully intentional.

Investigator Information:

Name- Jordan Shepherd

Experience- Digital Forensics Student

Tools Used- Linux, Autopsy, Wireshark

Investigator has continued to further knowledge as well as experience within various sleuth kit platforms. Deepening the pot of knowledge needed to carry out forensics cases successfully.

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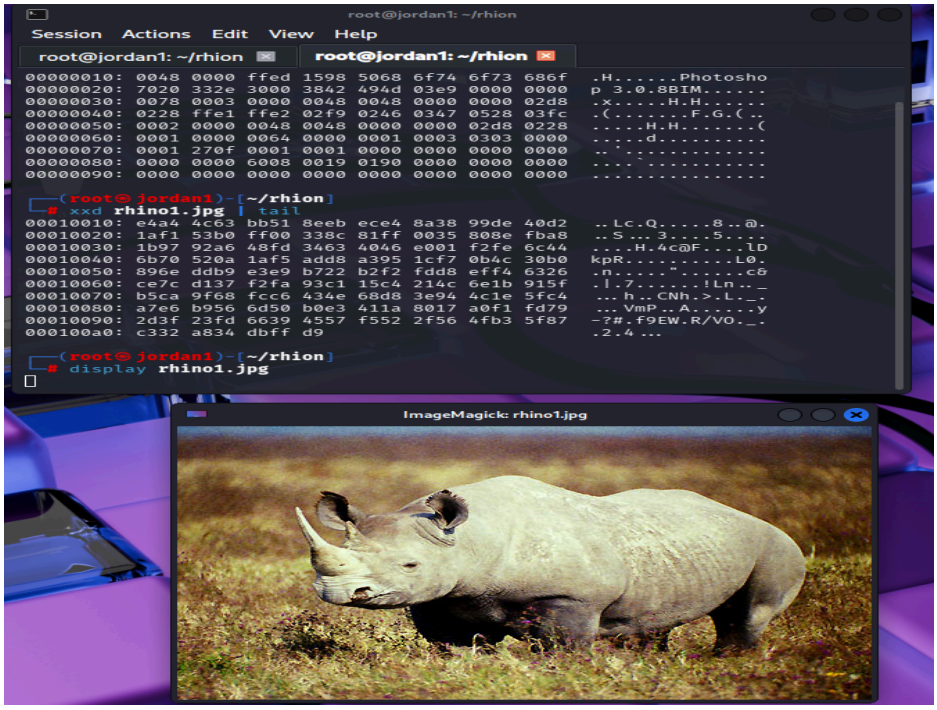
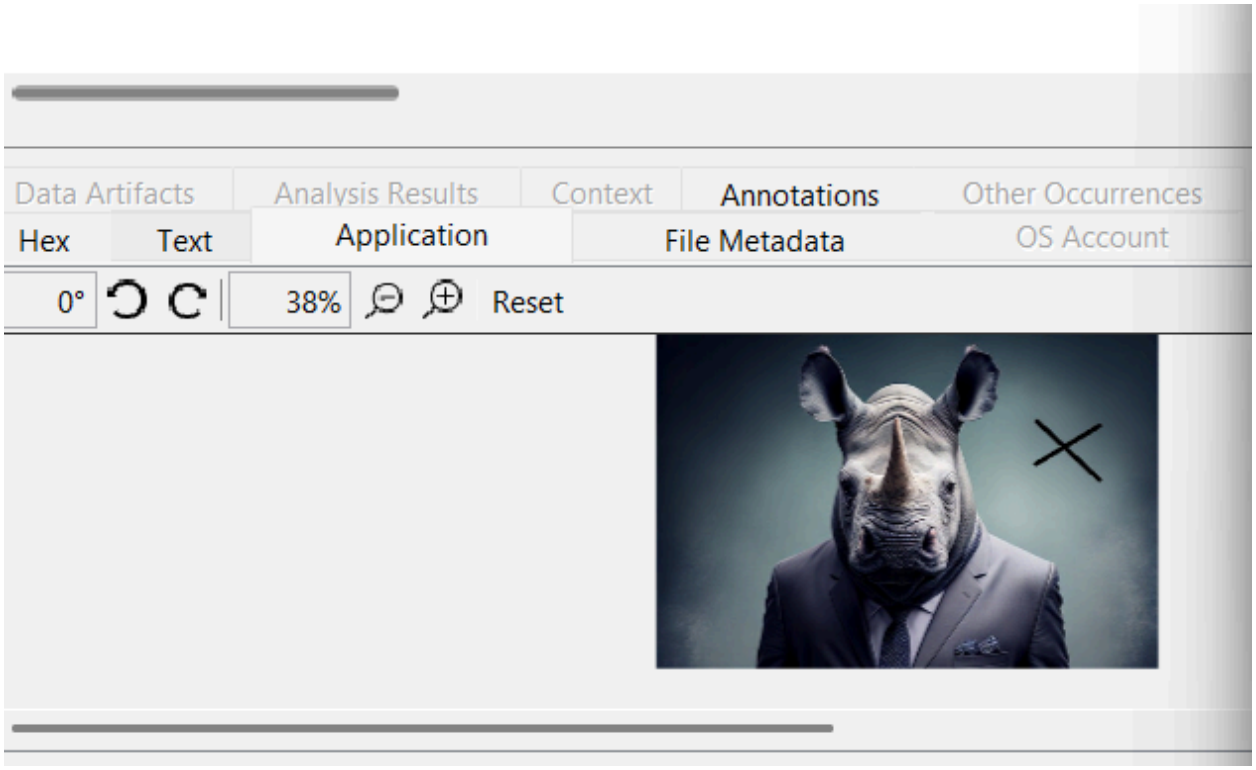
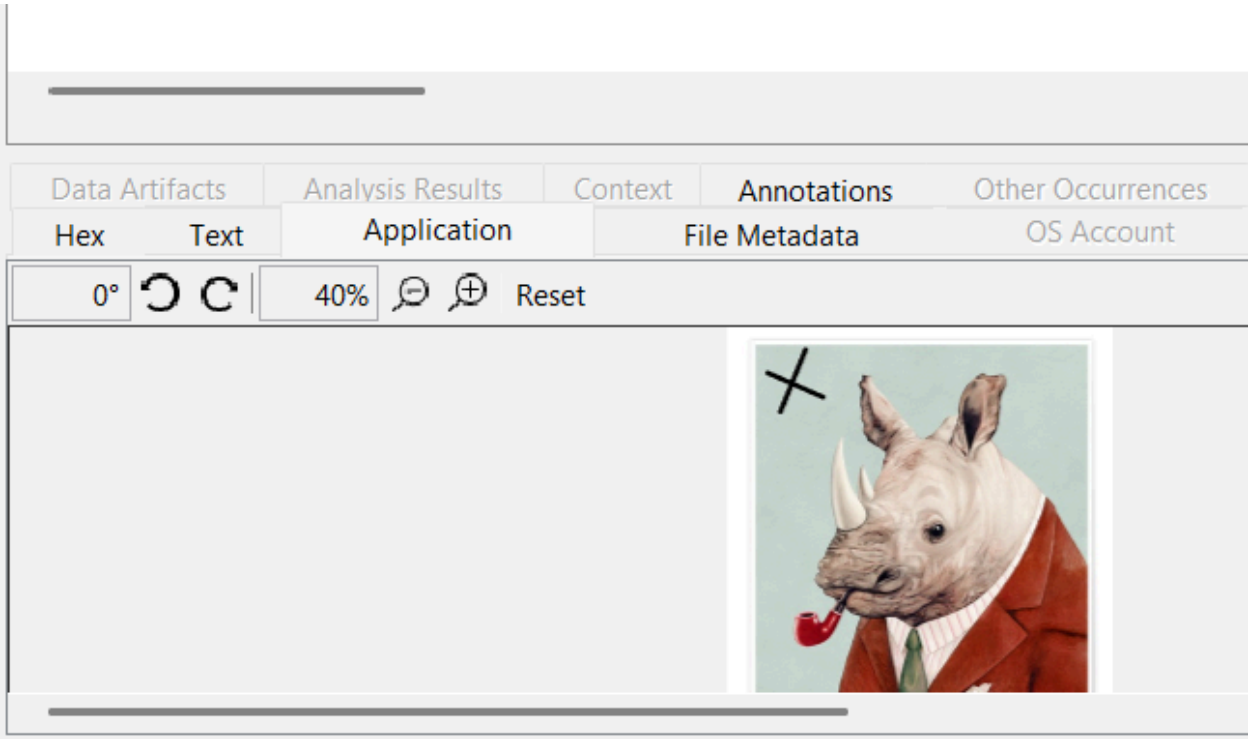
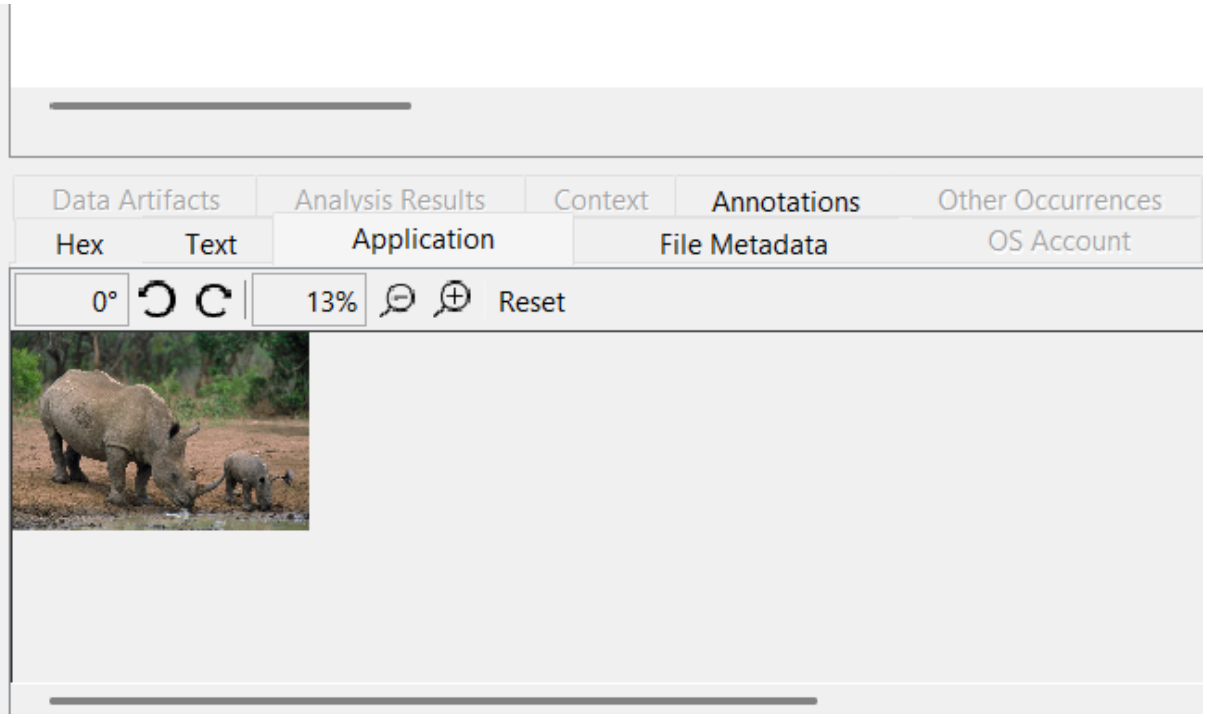
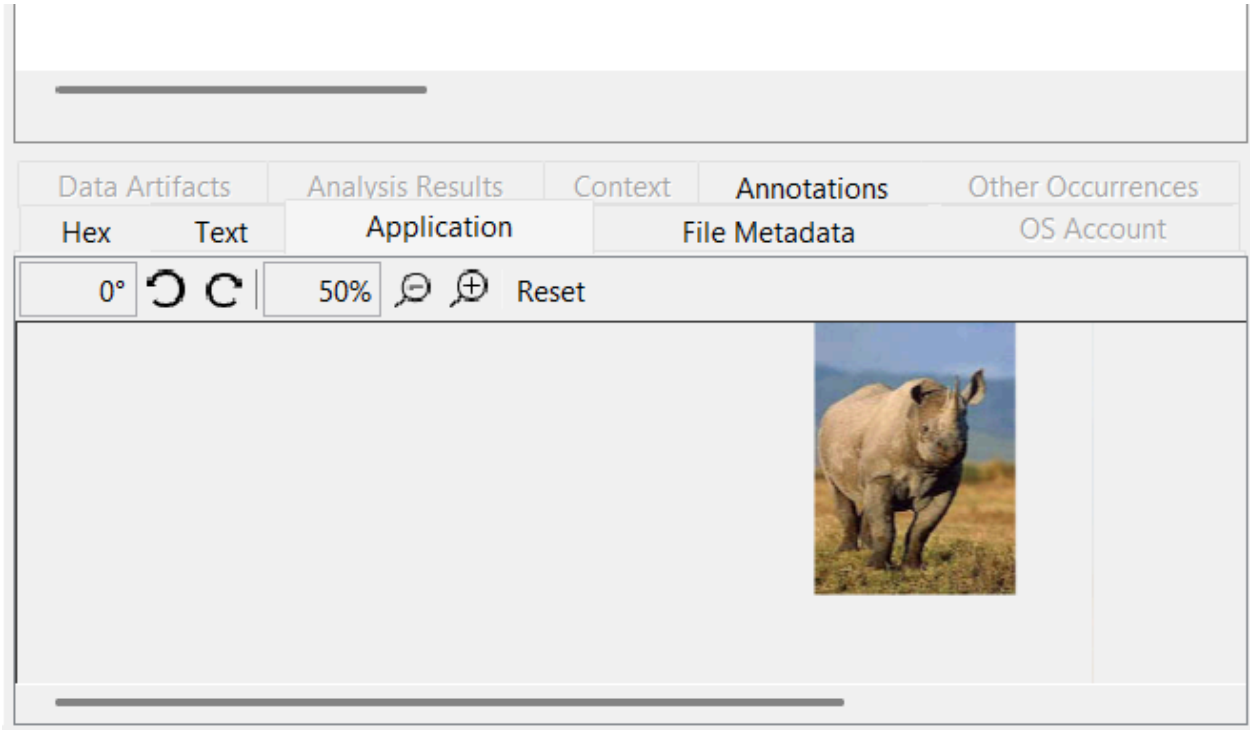
Suspect Actions (Rhino Images)



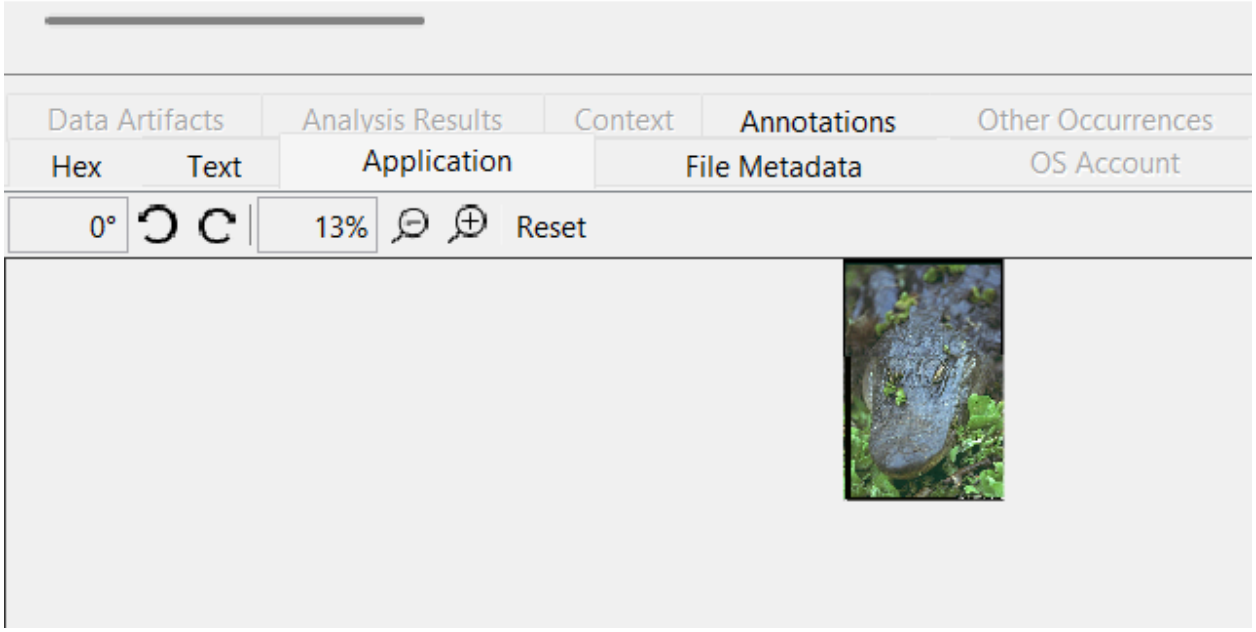

Data Artifacts
Analysis Results
Context
Annotations
Other Occurrences
Hex
Text
Application
File Metadata
OS Account
0°
38%
Reset

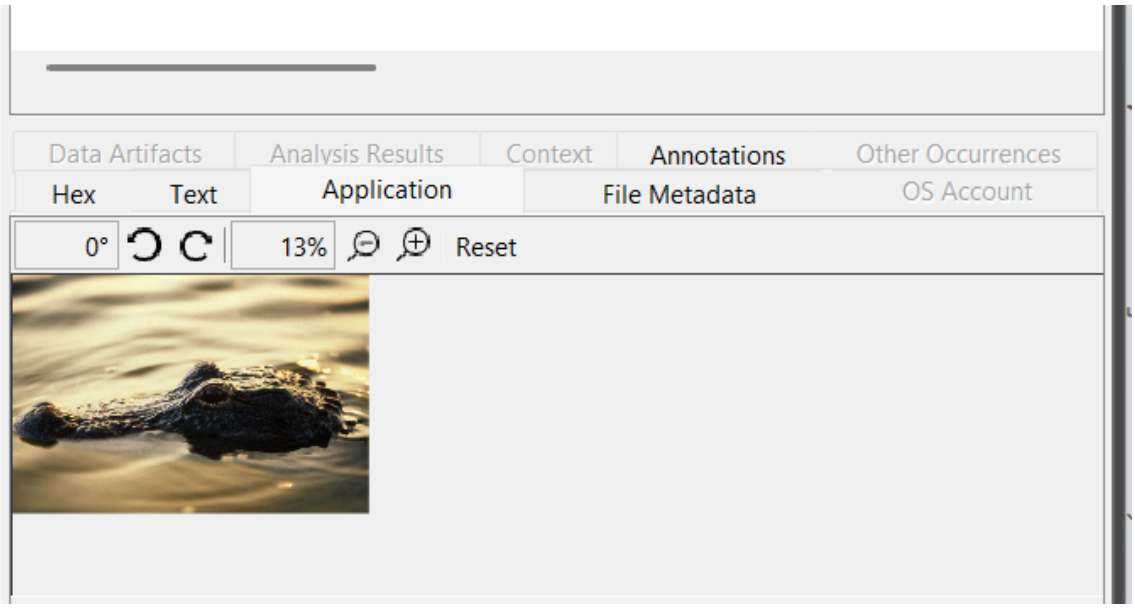

Data Artifacts
Analysis Results
Context
Annotations
Other Occurrences
Hex
Text
Application
File Metadata
OS Account
0°
40%
Reset

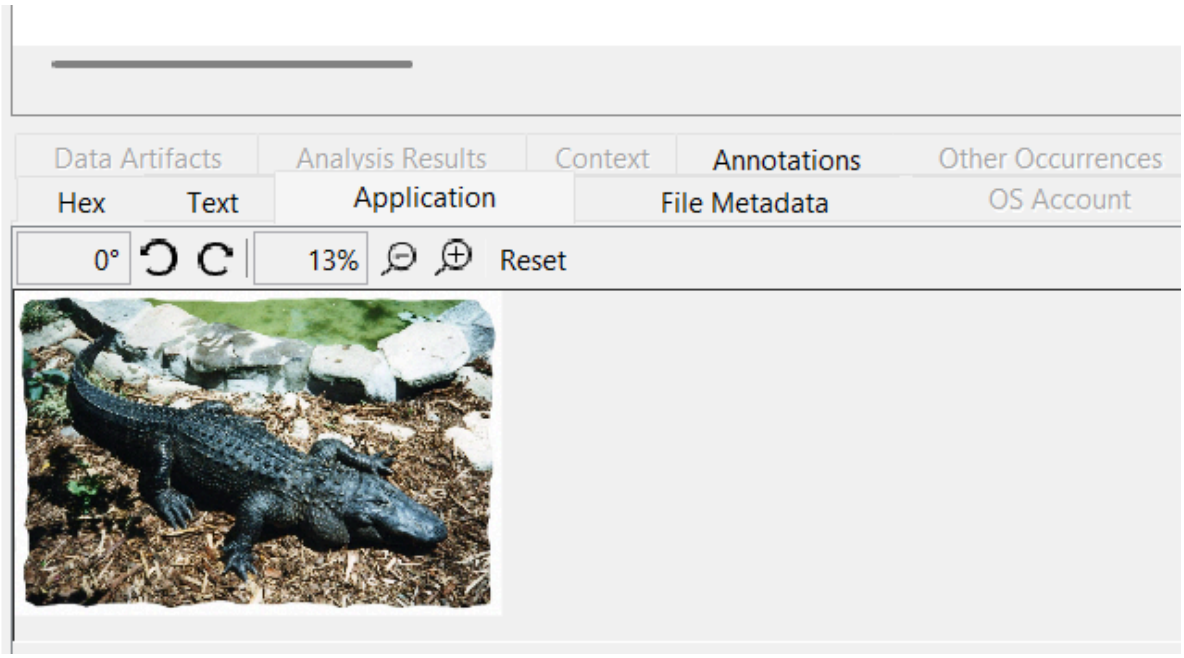

Data Artifacts
Analysis Results
Context
Annotations
Other Occurrences
Hex
Text
Application
File Metadata
OS Account
0°
13%
Reset

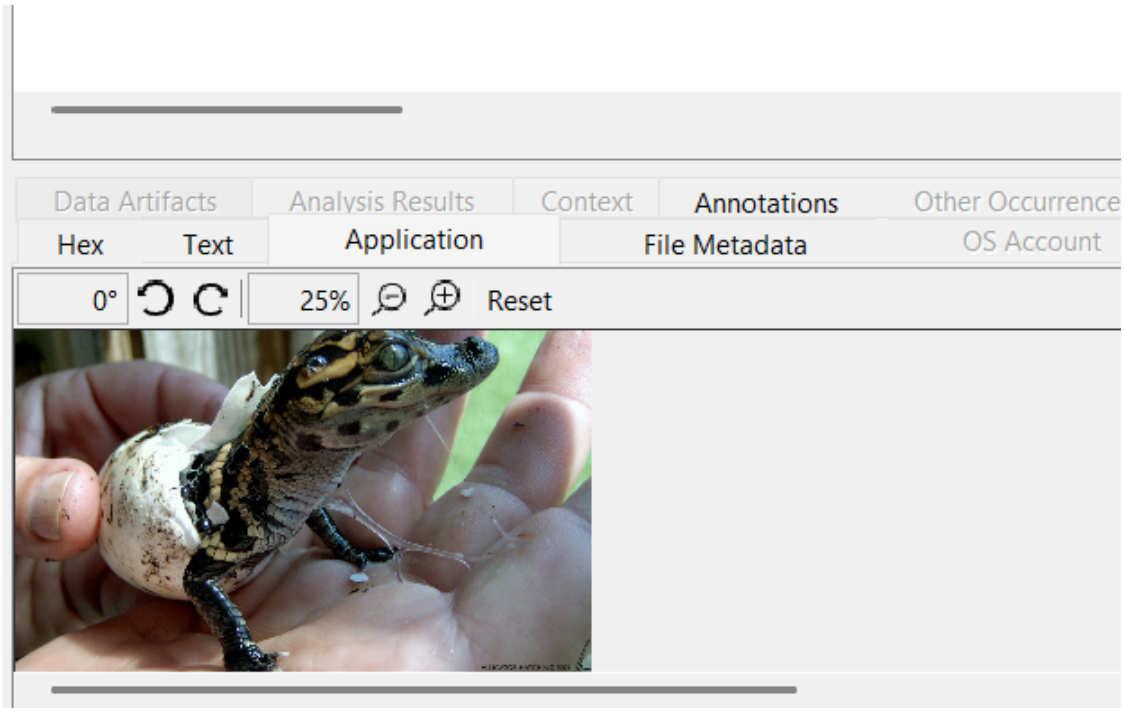

Data Artifacts
Analysis Results
Context
Annotations
Other Occurrences
Hex
Text
Application
File Metadata
OS Account
0°
50%
Reset

Suspect Images (Decoy Images)






Data Artifacts
Analysis Results
Context
Annotations
Other Occurrences
Hex
Text
Application
File Metadata
OS Account
0°
13%
Reset


Data Artifacts
Analysis Results
Context
Annotations
Other Occurrence
Hex
Text
Application
File Metadata
OS Account
0°
25%
Reset

Evidence:

-Logs from the file system that shows file access

-Metadata

-User activity timelines

Investigation Tools:

Wireshark- Wireshark is a free publicly available packet analyzer. Its applications include network troubleshooting, analysis, software and communication protocol creation, and education.

Autopsy- A file system analysis software that optimizes image viewing.

Data Transfer/Concealment:

-Secret or disguised files

-Unusual file types

-Log entries show compression or hiding.

-File data was destroyed.

Investigator Activity:

Within this case study, the following actions took place. Data was collected and documented throughout the chain of custody. Then, forensic images were created with a write blocker. Also, Hash values are verified to ensure their validity. Disk images were loaded into Autopsy/Wireshark, and Files (rhino.log, rhino2.log, rhino3.log) were examined. Recovered destroyed and concealed files were shown and the timeline was reconstructed with artifacts.

Conclusion:

This forensic investigation revealed very clear proof of insider data leakage actions. The suspect accessed sensitive data, moved it to external storage, and tried to hide and destroy evidence. Multiple artifacts, including logs and recovered files, retrieve these results.

Check List:

TASK 1:YES

TASK 2:YES

TASK 3:YES

TASK 4:YES

TASK 5:YES

HYPOTHESIS 1: X

HYPOTHESIS 2: X

HYPOTHESIS 3: X